Anumana (inference): a reliable instrument of knowledge (***pramana***) about something that is beyond the senses. It requires cognitions (***jnana***) arising from sensory perceptions. According to the ***Nyaya*** realism, it begins with a doubt, such as whether there is a fire on a remote mountain. The relevant observation is that we can see smoke. In this case, fire is technically called the ***sadhya*** – that which is to be established or the property to be proved. The mountain is called the subject (***paksha***) and the smoke is called the logical reason (***hetu***) in the inferential process. We already know that there is no smoke without fire (this invariable association is called ***vyapti***) from familiar instances such as the kitchen. By way of corroboration, we also know the truth of the contraposed version of the generalisation: 'no fire, no smoke' from cases like the lake. We apply knowledge of the general principle to the case in question and can safely conclude that there is indeed fire on the mountain although we do not see the fire.

A demonstrative inference (*prayoga*) used to persuade someone else (*parartha-anumana*) would be formulated by the ***Nyaya-Vaisheshikas*** as:

Statement of the position or uncertainty (*pratijna*): 'There is fire (***sadhya***) on the mountain (***paksha***).'
Logical reason (***hetu***): Because there is smoke on the mountain.
General principle (***vyapti***): 'Wherever there is smoke, there is fire' that is supported by evidence – like a kitchen (***sapaksha***); unlike a lake (***vipaksha***).'
Application (*upanaya*): 'There is smoke on the mountain', which states that the subject has the logical reason that is always associated with (pervaded by or included in – *vyapta*) the property to be proved.
Conclusion (*nigamana*): Therefore, there is fire.